Generative design of novel RNA-based life

Team: Eve

Conceptual Framework

Completed

EVA

Unified RNA generative model

modules

CRISPER

elements

mRNA

tRNA

ncRNA

mRNA

Planned

RNA genome generative model

EVA2

dsRNA
e.g. Reovirus

ssRNA
e.g. Poliovirus

saRNA vaccine

minimal self-replication ribozyme

RNA Life

RNA Virus
~4–40kb

Self-amplifying RNA
~0–15kb

RNA elements & modules
~0–8kb

EVA2

EVA1

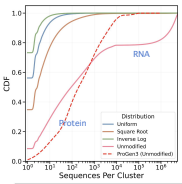
Methodology

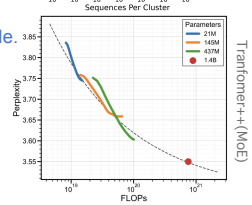
Findings in EVA1

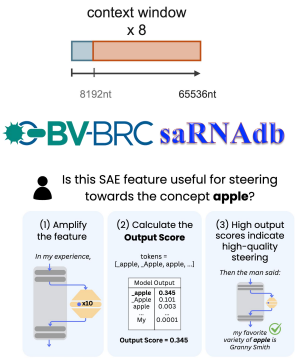
1. RNA is less evolutionarily conserved than protein and DNA.
   - Special sampling strategy and training skills are needed.

2. Better architecture is needed.
   - Transformer++ struggles to scale. (perplexity lower bound: 3.47)

3. Generative AI is interpretable.
   - Using SAE to Align neurons with biological features







Possible solutions in EVA2

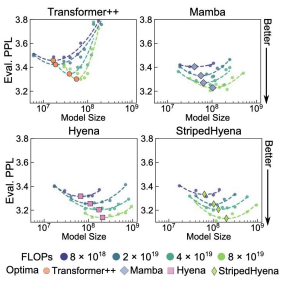
1. Better model architecutre for RNA
   - Lower perplexity (3.47--->?)
   - Better scaling (1.4B--->7B)
   - Longer window (8192nt--->65536nt)

2. saRNA & RNA virus data (Safety & Ethical)
   - VirusCircBase & Serratus & ViRBase & BV-BRC & UTRdb & saRNAdb & VirJenDB

3. Steering model for controllable design
   - From identifying features in models
     To steering features for function design

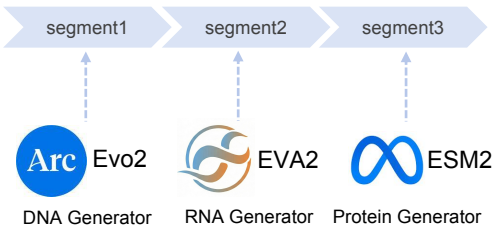
Output

Better architectures and the first scaling laws for RNA



the First AI-designed RNA-based Life

Unified Platform for Generative Biology



Needs

1. Compute resources
   - 32 H200 GPU for 3-4 month

2. Expert guidance to identify optimal RNA virus targets
   - Bioethical Compliance
   - Experimental Feasibility
   - Therapeutic Value

About us

- Autonomous Problem-Solver: High initiative in independent research and solving complex technical challenges.
- Full Commitment: Available for full-time, on-site engagement to ensure dedicated project execution.
- AI for Bio Enthusiast: Strong communicator passionate about AI and Generative Biology
- Vision: To synthesize the first AI-designed RNA life and secure our place in the history of Generative Biology